Extracted Equations

Formal relations and surrounding extracted text from Chapter 7.

No.	Name	Equation / Symbol	Extracted context	Page
7.1	Critical section structure	wait(S_k) ... signal(S_k)	each critical section operating on a resource R_k must begin with a wait(S_k) primitive and end with a signal(S_k) primitive.	1
7.1	Blocking time	B_i	B_i denotes the maximum blocking time task τ_i can experience.	5
7.1	Generic critical section	$z_{i,k}$	$z_{i,k}$ denotes a generic critical section of task τ_i guarded by semaphore S_k.	5
7.1	Longest critical section	$Z_{i,k}$	$Z_{i,k}$ denotes the longest critical section of task τ_i guarded by semaphore S_k.	5
7.1	Duration	$\delta_{i,k}$	$\delta_{i,k}$ denotes the duration of $Z_{i,k}$.	6
7.1	Blocking critical sections	$\gamma_{i,j} = \{Z_{j,k} \mid (P_j < P_i) \text{ and } (S_k \in \sigma_{i,j})\}$	$\gamma_{i,j}$ denotes the set of the longest critical sections that can block τ_i, accessed by the lower priority task τ_j.	6
7.2	NPP priority	$p_i(R_k) = \max_h \{P_h\}$	as soon as a task τ_i enters a resource R_k, its dynamic priority is raised to the level.	6
7.3	NPP blocking	$B_i = \max_{j,k} \{\delta_{j,k} - 1 \mid Z_{j,k} \in \gamma_i\}$	the maximum blocking time τ_i can suffer is given by the duration of the longest critical section among those belonging to lower priority tasks.	8
7.4	HLP priority	$p_i(R_k) = \max_h \{P_h \mid \tau_h \text{ uses } R_k\}$	as soon as a task τ_i enters a resource R_k, its dynamic priority is raised to the level.	8
7.5	Resource ceiling	$C(R_k) = \max_h \{P_h \mid \tau_h \text{ uses } R_k\}$	assigning each resource R_k a priority ceiling $C(R_k)$ (computed off-line) equal to the maximum priority of the tasks sharing R_k.	8
7.6	HLP blocking	$B_i = \max_{j,k} \{\delta_{j,k} - 1 \mid Z_{j,k} \in \gamma_i\}$	Since a task τ_i can be blocked at most once, the maximum blocking time τ_i can suffer is given by the duration of the longest critical section among those that can block τ_i.	10
7.7	PIP priority inheritance	$p_j(R_k) = \max\{P_j, \max_h \{P_h \mid \tau_h \text{ is blocked on } R_k\}\}$	In general, a task inherits the highest priority of the tasks it blocks.	11
7.8	PIP bound	$\alpha_i = \min(l_i, s_i)$	a task τ_i can be blocked for at most the duration of $\alpha_i = \min(l_i, s_i)$ critical sections.	15
7.9	PIP lower task sum	$B_i^l = \sum_{j:P_j<P_i} \max_k \{\delta_{j,k} - 1 \mid Z_{j,k} \in \gamma_i\}$	for each lower priority task τ_j that can block τ_i, sum the duration of the longest critical section among those that can block τ_i.	17
7.10	PIP semaphore sum	$B_i^s = \sum_{k=1}^{m} \max_j \{\delta_{j,k} - 1 \mid Z_{j,k} \in \gamma_i\}$	for each semaphore S_k that can block τ_i, sum the duration of the longest critical section among those that can block τ_i.	17
7.11	PIP upper bound	$B_i = \min(B_i^l, B_i^s)$	compute B_i as the minimum between B_i^l and B_i^s.	17
7.12	Semaphore ceiling	$C(S_k) = \max_i \{P_i \mid S_k \in \sigma_i\}$	for each semaphore S_k we define a ceiling $C(S_k)$ equal to the highest-priority of the tasks that use S_k.	17
7.13	PCP ceiling	$C(S_k) = \max_i \{P_i \mid S_k \in \sigma_i\}$	Each semaphore S_k is assigned a priority ceiling $C(S_k)$ equal to the highest priority of the tasks that can lock it.	23
7.14	PCP inheritance	$p_j(R_k) = \max\{P_j, \max_h \{P_h \mid \tau_h \text{ is blocked on } R_k\}\}$	In general, a task inherits the highest priority of the tasks blocked by it.	23
7.15	PCP blocking set	$\gamma_i = \{Z_{j,k} \mid (P_j < P_i) \text{ and } C(R_k) \geq P_i\}$	a task τ_i can only be blocked by critical sections belonging to lower priority tasks with a resource ceiling higher than or equal to P_i.	28